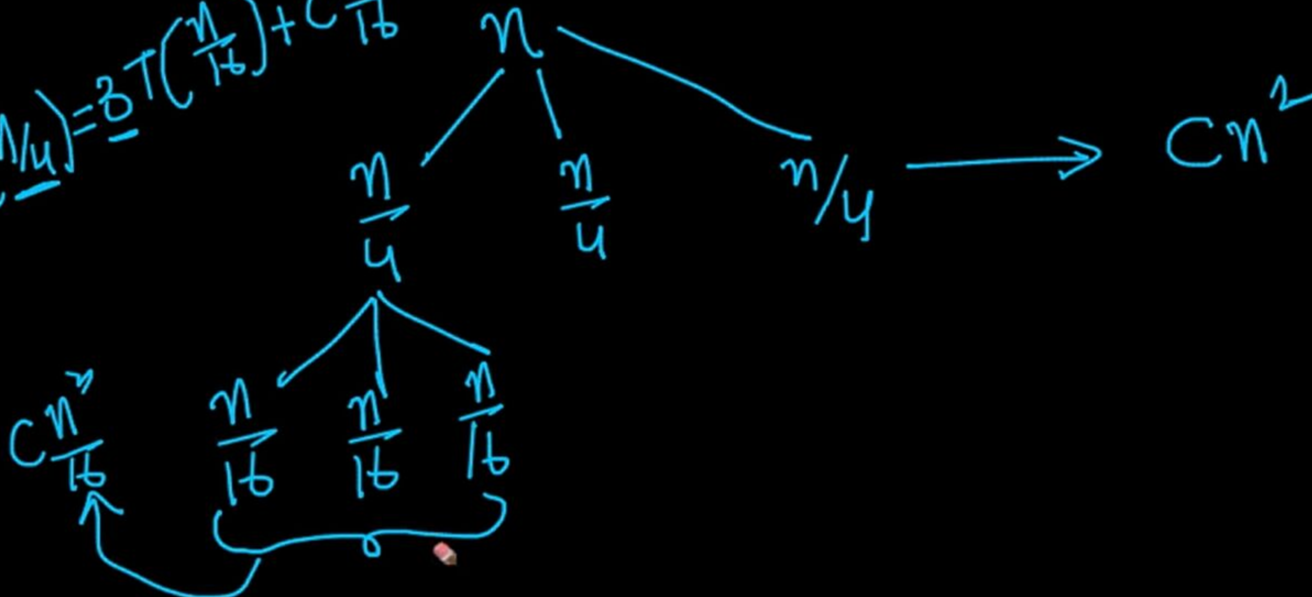
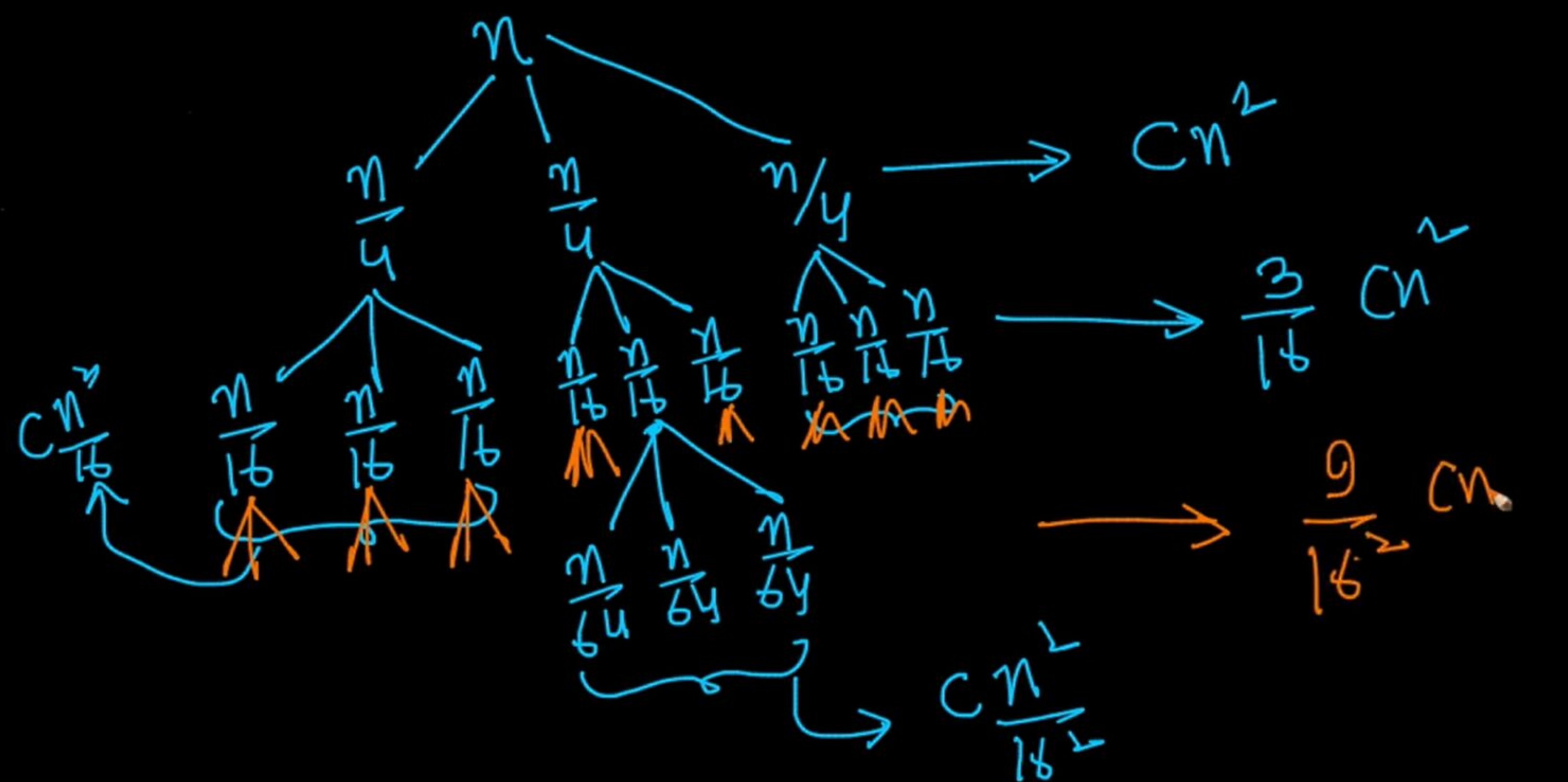
$$T(n) = \boxed{3}\,T\left(\lfloor n/4 \rfloor\right) + cn^2 \qquad \lfloor n/4 \rfloor = \frac{n}{4}$$

$$T(n/4) = 3T\left(\frac{n}{16}\right) + c\frac{n^2}{16}$$



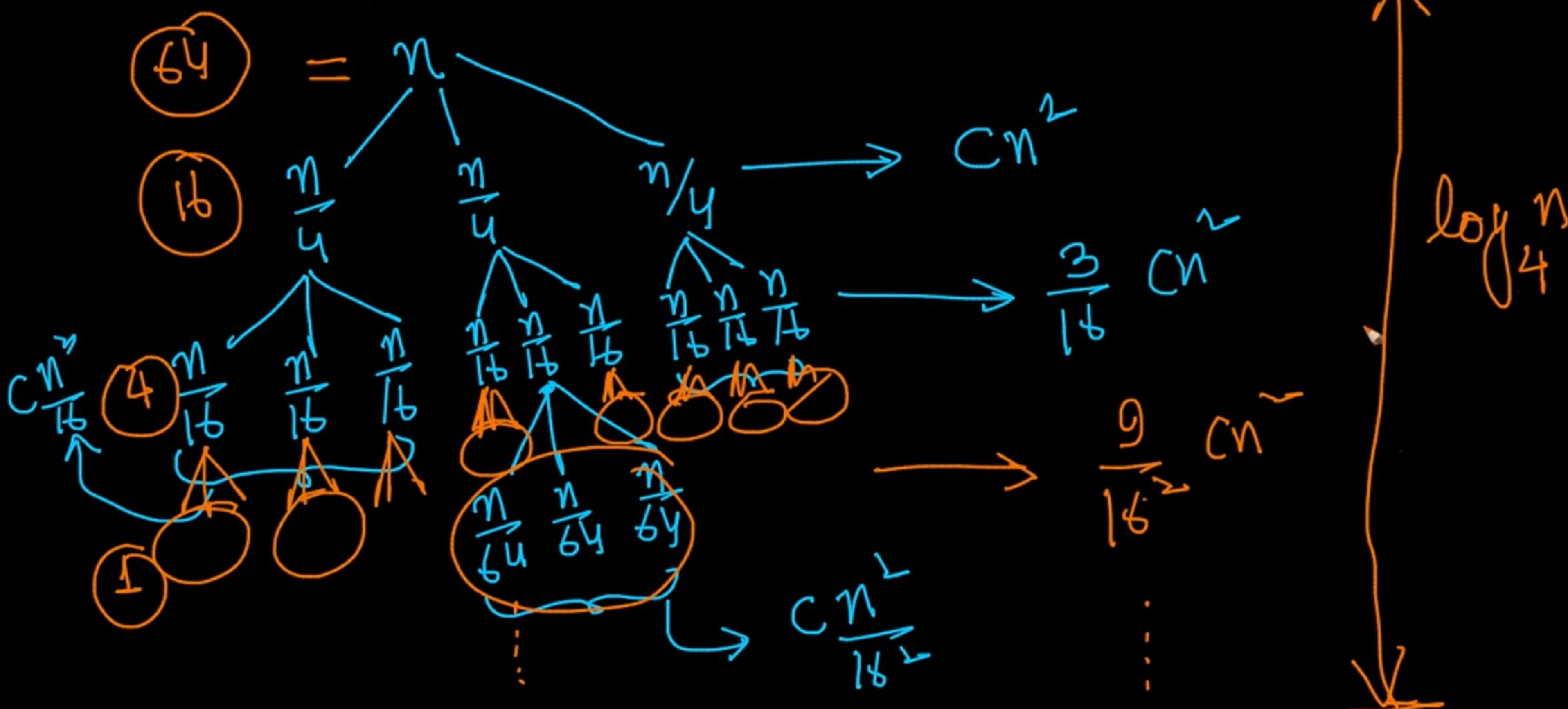
three problems again I take C n square
by 16 time okay very simple let

$$T(n) = \boxed{3}\,T(\lfloor n/4 \rfloor) + cn^2 \qquad \lfloor n/4 \rfloor = \frac{n}{4}$$



by 16 squared 1 2 3 4 5 6 7 8 9 so at
this stage I have 9 by 16 square CN

$$T(n) = \boxed{3}\,T(\lfloor n/4 \rfloor) + cn^2 \qquad \lfloor n/4 \rfloor = \frac{n}{4}$$

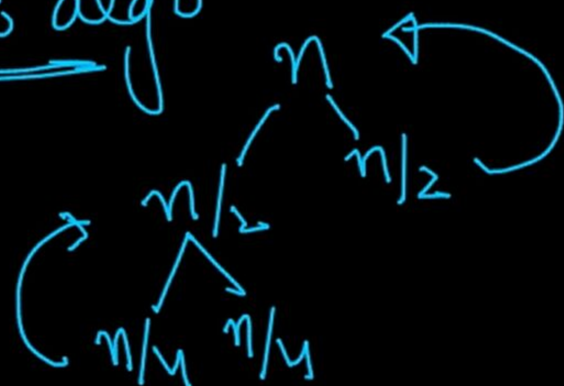


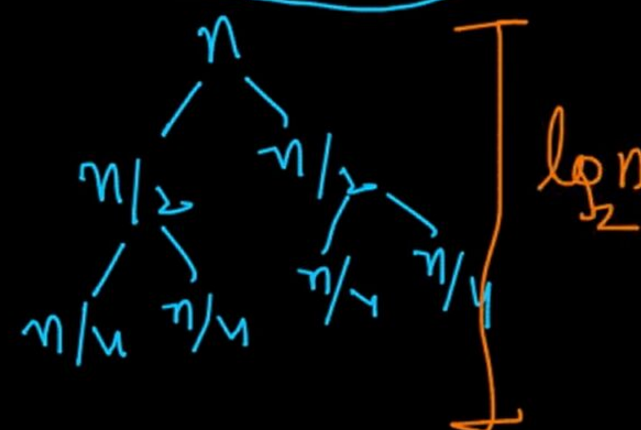
the depth of the tree is nothing but log
n base 4

Solving recurrences: Recursion Tree method

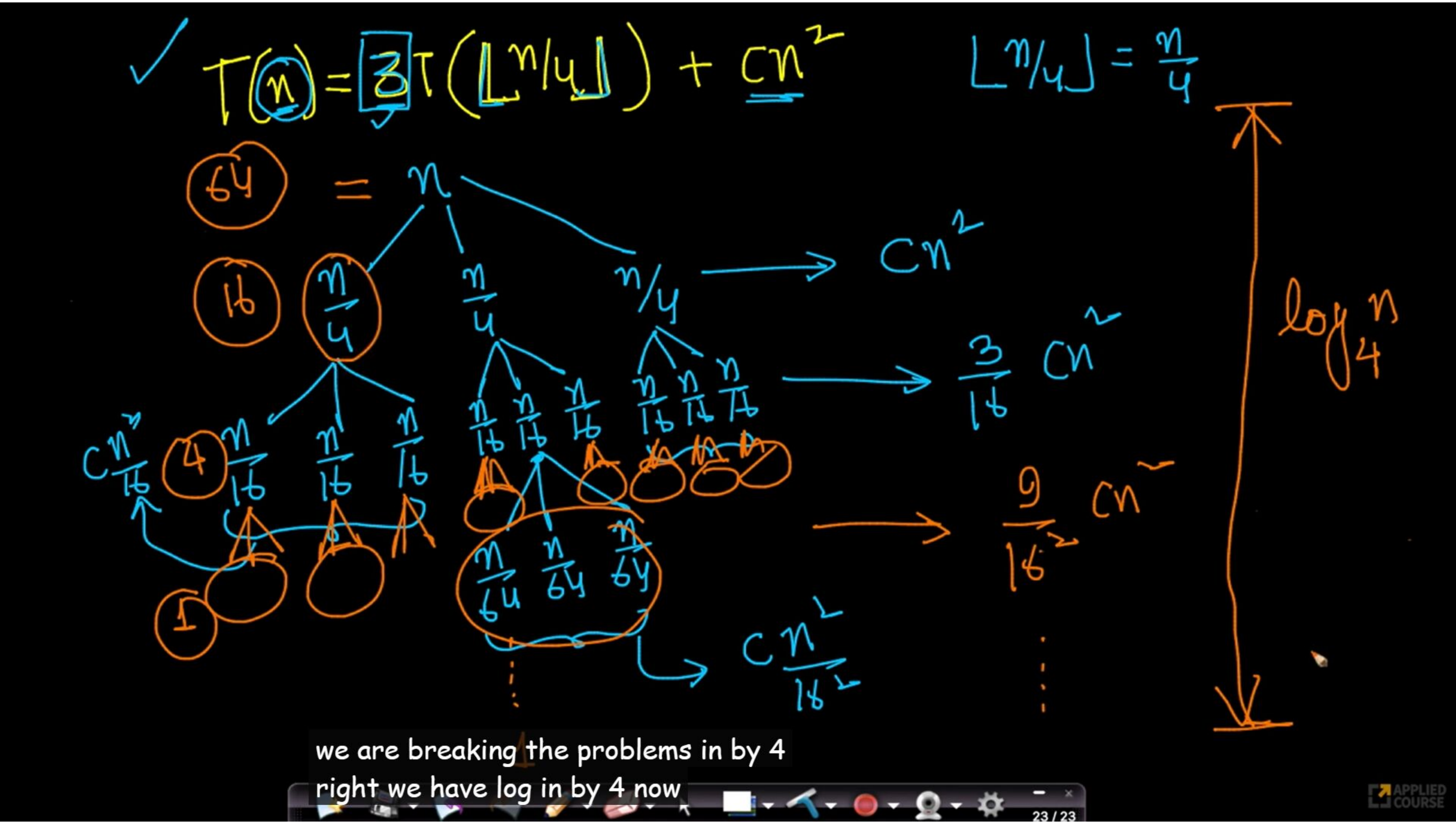
Merge Sort :- $T(n) = 2T(n/2) + cn$ ← recurrence relation

recursive algo





showed that that the depth of this tree
is log n base 2 similarly



T(n) = 3T(⌊n/4⌋) + cn^2
⌊n/4⌋ = n/4
cn^2
3/16 cn^2
9/16^2 cn^2
log_4 n
we are breaking the problems in by 4
right we have log in by 4 now
APPLIED COURSE
23 / 23

DS_Algo - Google Drive | Data Structures & Algorithms | Geometric series - Wikipedia | Master theorem (analysis of al

https://en.wikipedia.org/wiki/Geometric_series#Formula

As n goes to infinity, the absolute value of r must be less than one for the series to converge. The sum then becomes

$$a + ar + ar^2 + ar^3 + ar^4 + \cdots = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \text{ for } |r| < 1.$$

When $a = 1$, this can be simplified to

$$1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r},$$

the left-hand side being a geometric series with common ratio r.

The formula also holds for complex r, with the corresponding restriction, the modulus of r is strictly less than one.

Proof of convergence [edit]

We can prove that the geometric series converges using the sum formula for a geometric progression:

$$1 + r + r^2 + r^3 + \cdots = \lim_{n\to\infty} \left(1 + r + r^2 + \cdots + r^n\right)$$

$$1 - r^{n+1}$$

geometric series right this is called a
geometric series okay each

27 / 27

DS_Algo - Google Drive | Data Structures & Algorithms | Geometric series - Wikipedia | Master theorem (analysis of al

https://en.wikipedia.org/wiki/Geometric_series#Formula

As n goes to infinity, the absolute value of r must be less than one for the series to converge. The sum then becomes

$$a + ar + ar^2 + ar^3 + ar^4 + \cdots = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \text{ for } |r| < 1.$$

When $a = 1$, this can be simplified to

$$1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r},$$

the left-hand side being a geometric series with common ratio r.

The formula also holds for complex r, with the corresponding restriction, the modulus of r is strictly less than one.

Proof of convergence [edit]

We can prove that the geometric series converges using the sum formula for a geometric progression:

$$1 + r + r^2 + r^3 + \cdots = \lim_{n\to\infty} \left(1 + r + r^2 + \cdots + r^n\right)$$

$$1 - r^{n+1}$$

is true if and only if the absolute value of R is less than 1 right

27 / 27

$$cn^2 + \frac{3}{16}cn^2 + \left(\frac{3}{16}\right)^2 cn^2 + \left(\frac{3}{16}\right)^3 cn^2 + \left(\frac{3}{16}\right)^4 cn^2 + \cdots$$

$$= cn^2 \left\{ \underbrace{1 + \overset{r}{\frac{3}{16}} + \overset{r^2}{\left(\frac{3}{16}\right)^2} + \overset{r^3}{\left(\frac{3}{16}\right)^3} + \overset{r^4}{\left(\frac{3}{16}\right)^4} + \cdots}_{\text{geometric progression}} \right\}$$

$$r = \frac{3}{16} < 1$$

$$cn^2 \left\{ \frac{1}{1 - \frac{3}{16}} \right\} = cn^2 \left\{ \frac{16}{13} \right\}$$

it right this is basically a constant 16
minus 3 is 13

$$T(n) = \cancel{c}n^2 + \cancel{\frac{1}{3}}$$

$$\boxed{T(n) = \Theta(n^2)}$$

solved right so I have derived that TN
is equal to order